

1. For the mano machine we discussed in class, assume we want to replace the AND instruction with a SUB instruction, whose function will be:

(a) $AC \leftarrow AC - M[\text{addr}]$

Write the complete instruction cycle analysis by specifying microoperations clock cycle by clock cycle along with its condition. (10 pts)

T0: $AR \leftarrow PC$

T1: $IR \leftarrow M[AR]$, $PC \leftarrow PC + 1$

T2: $I \leftarrow IR(15)$, $AR \leftarrow IR(0-11)$, $D7...D0 \leftarrow \text{Decode}(IR(12-14))$

T3 · I · D7': $AR \leftarrow M[AR]$

T4 · D1: $DR \leftarrow M[AR]$

T5 · D1: $AC \leftarrow AC - DR$, $E \leftarrow C_{\text{out}}$, $SC \leftarrow 0$

- (b) What if we want to replace the AND instruction with a SEQ instruction. SEQ is defined as: SEQ: if ($M[AR] == AC$), skip next instruction. Do the same instruction analysis to the SEQ.(10pts)

T0: $AR \leftarrow PC$

T1: $IR \leftarrow M[AR]$, $PC \leftarrow PC + 1$

T2: $I \leftarrow IR(15)$, $AR \leftarrow IR(0-11)$, $D7...D0 \leftarrow \text{Decode}(IR(12-14))$

T3 · I · D7': $AR \leftarrow M[AR]$

T4 · D0: $DR \leftarrow M[AR]$

T5 · D0 · ($AC - DR = 0$): $PC \leftarrow PC + 1$, $SC \leftarrow 0$

T5 · D0 · ($AC - DR \neq 0$): $SC \leftarrow 0$

2. The operations to be performed with a flip-flop F are specified by the following register transfer statement:

xT3: $F \leftarrow 1$; set F

yT1: $F \leftarrow 0$; clear F

zT2: $F \leftarrow \neg F$; complement F

wT5: $F \leftarrow G$; load G

Implement with a JK flip-flop (10 pts)

$J = xT3 \vee zT2 \vee (wT5 \wedge G)$

$K = yT1 \vee zT2 \vee (wT5 \wedge \neg G)$